ARGHADEEP FADIKAR

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PROFILE SUMMARY

IIT Alumnus and Data Scientist with nearly 2 years of experience building **production-grade Industrial AI solutions** using **Computer Vision, NLP, Deep Learning, and Agentic AI**. Specialized in **Object detection, OCR, self-supervised learning, Unsupervised Learning and multi-modal AI** systems using **YOLO, DINO, JEPA, World Models, LLMs, Vision Language Models and Transformer-based architectures**. Experienced in developing and deploying scalable real-time AI pipelines **with FastAPI, Docker, Kubernetes, Redis, PostgreSQL and GPU accelerated workflows** for manufacturing analytics, predictive maintenance and automated quality inspection systems.

ORGANISATIONAL EXPERIENCE

Data Scientist , Michelin *[June,2024-current]*

Project: Defect Label Ambiguity Detection

- Engineered and deployed a robust **relabeling framework and algorithm** leveraging image feature correlations and label distribution patterns to resolve label inconsistencies in noisy datasets and **enhance** defect localization and classification **accuracy by 75%.**
- Increased operational efficiency , manufacturing quality by faster root cause analysis, **minimizing manual relabeling effort** and improving trustworthiness of AI-driven inspection systems, contributing to **reduced quality control costs** and higher process reliability.
- Successfully adopted across **multiple Michelin manufacturing factories**, enabling standardized and scalable dataset quality improvement workflows for industrial AI systems and also this algorithm is widely used in multiple use cases related to computer vision.

Project: Special Marking Detection & Text Recognition

- Developed and deployed a high-performance **Computer Vision pipeline** integrating advanced **Object Detection and OCR models**, optimized for efficient deployment on **edge devices** with **low-latency** real-time inference in industrial production environments
- Developed an in-house **OCR architecture from scratch** for tire image text recognition, delivering high accuracy, **low-latency edge deployment**, easy retraining capabilities, and **reduced dependency** on external paid OCR solutions.
- Enabled precise identification of tire logos and extraction of textual markings with **~99%** accuracy and optimized system latency for production.
- **Integrated the pipeline** in industrial camera systems near conveyor belts and inspection stations across multiple factories to enable automated tire marking verification , continuous real-time monitoring and improved traceability within production workflows.

Project: Self-Supervised Learning for Defect Detection

- Built a robust **self-supervised learning framework** to **reduce dependency** on large-scale labeled datasets for industrial defect detection.
- Applied **Auto Encoder, DINO , JEPA and LoRA-based fine-tuning** to achieve strong detection performance with minimal annotations.
- **Lowered data labeling effort by 80%** while maintaining high model efficiency, adaptability, and scalability across diverse defect scenarios.
- Developed a generalized **self-supervised backbone model** for tire industry, **trained from scratch** using self supervised learning on industrial tire data to support multiple downstream tasks **with minimal amount of labeling** effort and rapid adaptation.
- **Accelerated AI model development cycles** by significantly reducing annotation overhead, enabling faster experimentation, rapid prototyping, and efficient deployment of scalable defect detection solutions across multiple industrial manufacturing use cases.
- **Successfully deployed** and actively utilized across multiple tire manufacturing factories, enabling scalable AI-driven inspection workflows, improving consistency in defect analysis, and supporting reliable real-time quality assurance in production environments.

Project: Tire Vision AI|Agentic AI Platform

- Developed and deployed a production-grade **multi-modal Agentic AI** system for tire defect detection, industrial safety monitoring, and predictive maintenance by integrating **Computer Vision and NLP** to enable intelligent real-time monitoring and automated decision support.
- Built autonomous AI tools using **YOLO, OpenCV, LangGraph, and FastAPI** for real-time anomaly detection and intelligent incident reporting.
- Designed scalable AI infrastructure using **Docker, Kubernetes, Redis, and PostgreSQL** for **low-latency real-time** manufacturing analytics.
- Improved operational efficiency and reduced manual monitoring effort by enabling AI-driven inspection and safety monitoring workflows.
- Enhanced production reliability and quality control through **faster incident response**, and **predictive maintenance capabilities**.

INTERNSHIP

Data Science Intern , Schlumberger *[May,2023-July,2023]*

- Conducted an in-depth benchmarking analysis comparing **YOLO-NAS** variants with **YOLOv5** across domain-specific object detection use cases.
- Identified YOLO-NAS-L as the top-performing model, **replacing the previously used YOLOv5** by achieving an **F1 score of 0.944** with nearly **2× faster inference** throughput, significantly **reducing video inference lag** in CCTV-based real-time monitoring systems.

TECHNICAL SKILLS

- Statistics & Probability
- Computer Vision
- GenAI
- Agentic AI
- Forecasting
- NLP
- Deep Learning
- Neural Network

CERTIFICATIONS

- Neural Networks & Deep Learning from Coursera
- Machine Learning from Coursera
- Unsupervised Learning, Recommenders, Reinforcement Learning from Coursera
- SQL from DataCamp

ACADEMIC DETAILS

- **M.Sc. in Applied Statistics & Informatics** from Indian Institute Of Technology , Bombay with 8.7 CPI in 2024
- **B.Sc. in Statistics** from Ramakrishna Mission Residential College , Narendrapur with 9.6 CPI in 2022

ACADEMIC ACHIEVEMENTS

- Awarded two **Gold Medals** for securing **First Rank** in the B.Sc. and for achieving the **Best Academic Performance** in the Science stream.
- Secured **All India Rank 22** in IIT JAM Mathematical Statistics Examination and **All India Rank 26** ISI MSTAT Examination.
- Earned the prestigious **Pfizer Scholarship at IIT Bombay** in recognition of exceptional academic performance.